The (r+1)th Term of a Binomial Expansion

The $(r+1)$th term of the binomial expansion of $(x+y)^n$ is:

$$\binom{n}{r}x^{n-r}y^r$$

How To

Given a binomial, write a specific term without fully expanding.

1. Determine the value of n according to the exponent.
2. Determine $(r+1)$.
3. Determine r.
4. Replace r in the formula for the $(r+1)$th term of the binomial expansion.

Example 3

Writing a Given Term of a Binomial Expansion

Find the tenth term of $(x+2y)^{16}$ without fully expanding the binomial.

Solution

Because we are looking for the tenth term, $r+1=10,$ we will use $r=9$ in our calculations.

$$\binom{n}{r}x^{n-r}y^r$$

$$\binom{16}{9}x^{16-9}(2y)^9 = 5{,}857{,}280x^7y^9$$

Try It #3

Find the sixth term of $(3x-y)^9$ without fully expanding the binomial.

Media

Access these online resources for additional instruction and practice with binomial expansion.

- The Binomial Theorem
- Binomial Theorem Example

9.6 Section Exercises

Verbal

1.
What is a binomial coefficient, and how it is calculated?